AWS Machine Learning Services - SAA-C03 Study Note

For the **AWS Certified Solutions Architect - Associate (SAA-C03)** exam, you generally do not need to know how to *code* these services. Instead, focus on **identifying the right service for a specific business problem**.

Key Concept: Most of these are "Serverless" and "Managed Services" (High Availability is built-in).

1. Amazon Rekognition

"Computer Vision / Image & Video Analysis"

- **What it does:** Uses ML to analyze images and videos. You send it an image; it tells you what is inside.
- **Key Features:**
  - **Object & Scene Detection:** Identifies cars, pets, beaches, cities.
  - **Facial Analysis:** Detects faces, gender, age range, emotions (happy/sad).
  - **Facial Recognition:** Matches a face against a private collection (e.g., employee badge entry).
  - **Content Moderation:** Automatically detects unsafe, offensive, or inappropriate content (nudity, violence).
- **Exam Scenario:**
  - "A company needs to automatically flag inappropriate images uploaded by users to a social media app." -> **Use Rekognition Content Moderation.**
  - "A store wants to track the path of customers walking through aisles." -> **Use Rekognition Video Pathing.**

2. Amazon Transcribe

"Speech-to-Text"

- **What it does:** Converts audio (MP3, WAV, etc.) into text.
- **Key Features:**
  - **Speaker Identification:** Can tell "Speaker A" from "Speaker B".
  - **Automatic Language Identification:** Detects the dominant language.
- **Exam Scenario:**
  - "A company needs to generate subtitles/captions for their video content automatically." -> **Use Transcribe.**
  - "You need to create searchable text transcripts from customer service call recordings." -> **Use Transcribe.**

3. Amazon Polly

"Text-to-Speech"

- **What it does:** Converts text into lifelike speech (MP3). It is the opposite of Transcribe.
- **Key Features:**
  - **Neural TTS:** Higher quality, more natural sounding voices.
  - **Lexicons:** Customize pronunciation of specific words (e.g., "AWS" as "A-W-S" instead of "Aws").
  - **SSML (Speech Synthesis Markup Language):** Markup tags to control breathing, volume, speed, and emphasis.
- **Exam Scenario:**
  - "An app needs to read articles aloud to visually impaired users." -> **Use Polly.**

4. Amazon Translate

"Language Translation"

- **What it does:** Provides high-quality neural machine translation between languages.
- **Key Features:**
  - Real-time and Batch translation.
  - Supports custom terminology (e.g., ensuring brand names aren't translated literally).
- **Exam Scenario:**
  - "A company wants to localize their website content into 10 different languages dynamically." -> **Use Translate.**

5. Amazon Lex + Amazon Connect

"Chatbots & Call Centers"

Amazon Lex (The "Brain")

- **What it does:** Provides Automatic Speech Recognition (ASR) and Natural Language Understanding (NLU). It is the technology behind **Alexa**.
- **Use Case:** Building conversational interfaces (Chatbots) for text or voice.

Amazon Connect (The "Phone System")

- **What it does:** A cloud-based **Contact Center** as a Service.
- **Integration:** You use **Connect** to handle the phone lines and routing, and you use **Lex** to build the automated "IVR" (Interactive Voice Response) that talks to the customer before an agent picks up.
- **Exam Scenario:**

- "A company needs a cloud-based call center that can handle spikes in traffic and uses AI to answer common questions." -> **Use Amazon Connect integrated with Amazon Lex.**

6. Amazon Comprehend

"Natural Language Processing (NLP)"

- **What it does:** Analyzes text to understand the *meaning* and *relationships* within it.
- **Key Features:**
  - **Sentiment Analysis:** Determines if text is Positive, Negative, Neutral, or Mixed.
  - **Entity Recognition:** Extracts people, places, brands, dates.
  - **PII Detection:** Detects Personally Identifiable Information (emails, SSNs) for redaction.
- **Exam Scenario:**
  - "A company wants to analyze customer emails to determine if they are happy or angry." -> **Use Comprehend Sentiment Analysis.**

7. Amazon Comprehend Medical

"NLP for Healthcare"

- **What it does:** Specifically trained on medical text to identify medical conditions, medications, dosages, and Protected Health Information (PHI).
- **Key Features:**
  - **HIPAA Eligible.**
  - Detects PHI (Patient Health Information) for anonymization.
- **Exam Scenario:**
  - "A hospital needs to index doctor's notes and identify patient diagnosis and prescribed medication." -> **Use Comprehend Medical.** (Do not choose standard Comprehend).

8. Amazon SageMaker

"ML for Developers / Custom Models"

- **What it does:** A fully managed service to build, train, and deploy **custom** machine learning models.
- **Key Concept:** Used when the "ready-made" AI services (like Rekognition or Polly) are not enough and you need to build your own algorithm.
- **Exam Scenario:**
  - "A data science team needs a managed environment to run Jupyter Notebooks and train TensorFlow models." -> **Use SageMaker.**

- "You need to deploy a custom fraud detection model with auto-scaling." -> **Use SageMaker.**

9. Amazon Kendra

"Enterprise Search"

- **What it does:** An intelligent document search service powered by ML. Think of it as "Google Search" for your internal company documents.
- **Key Features:**
  - Extracts answers from PDFs, Word docs, HTML, SharePoint, etc.
  - Supports **Natural Language Queries** (e.g., "Where is the IT support desk?" vs. matching keyword "IT").
- **Exam Scenario:**
  - "A company wants to allow employees to search through an internal repository of PDF manuals using natural language questions." -> **Use Kendra.**

10. Amazon Personalize

"Real-time Recommendations"

- **What it does:** Machine learning service for creating real-time personalized recommendations. Same technology used by Amazon.com for "Items you might like".
- **Exam Scenario:**
  - "An e-commerce website wants to show a 'Recommended for you' section based on user purchase history." -> **Use Personalize.** (Do not build a custom model in SageMaker if Personalize can do it).

11. Amazon Textract

"OCR on Steroids / Document Analysis"

- **What it does:** Extracts text, handwriting, and **data** from scanned documents.
- **Key Difference vs. OCR:** Standard OCR just gives you a blob of text. Textract understands **Structure** (Tables, Forms, Key-Value pairs).
- **Exam Scenario:**
  - "A company needs to extract total amounts and vendor names from thousands of scanned PDF invoices and load them into a database." -> **Use Textract.**

Machine Learning Summary Cheat Sheet

Service	Keyword / Trigger
Rekognition	Images, Video, Faces, Content Moderation (Safety), Labels.
Transcribe	Speech-to-Text, Subtitles, Meeting minutes, Audio files to text.

Service	Keyword / Trigger
Polly	Text-to-Speech, Talking app, SSML, Voices.
Translate	Localization, Language Translation.
Lex	Chatbots, Conversational AI, Alexa tech.
Connect	Cloud Contact Center, Call Center, Phone system.
Comprehend	Sentiment Analysis, NLP, Entities, PII redaction.
SageMaker	Custom Models, Jupyter Notebooks, Build/Train/Deploy own algorithm.
Kendra	Internal Search, Document Search, Natural Language Queries.
Personalize	Recommendations, "Users who bought this also bought...", Retail ML.
Textract	Scanned Documents, Forms, Tables, Invoices, Receipts, OCR.